

Week 10 - Radiosondes: Practice Questions

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

Attempt the questions before looking at the worked answers. Show equations, substitutions and units for numerical work, and use sketches where they help. You are **not** expected to complete every Extra Challenge problem. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

The table below is a synthetic radiosonde profile designed for practice. Wind direction is the direction *from* which the wind blows. Treat the measurements as exact unless the question asks about uncertainty.

Height (m)	Pressure (hPa)	T ($^{\circ}\text{C}$)	RH (%)	Wind dir. ($^{\circ}$)	Speed (m s^{-1})
0	1010	14	75	220	4
500	955	10.5	82	230	6
1000	900	7	91	240	8
1500	845	3.5	97	250	10
2000	795	0	93	260	12
3000	700	-6	65	270	16
5000	540	-19	40	280	22
8000	360	-40	30	290	30
10000	265	-54	25	300	38
12000	195	-53	20	310	42

YOU SHOULD TRY Question 1: Which quantities are measured directly and which are derived?

Using the columns in the sounding table:

1. Identify the quantities measured by sensors carried on the radiosonde.
2. Explain how wind speed and direction are obtained during the flight.
3. Height is often calculated from pressure/GPS information rather than measured by a mechanical “height sensor”. State one reason why it is useful to report the profile versus both height and pressure.

YOU SHOULD TRY Question 2: Diagnose the thermodynamic structure of the sounding

Use the table to answer all parts.

1. Calculate the mean environmental lapse rate from the surface to 2.0 km.
2. Compare it with the DALR, 9.8 K km^{-1} , and state whether an unsaturated parcel is dry-statically stable through this layer.
3. Identify the layer most likely to contain cloud from the RH observations.
4. Estimate the freezing level.
5. Identify the main dry layer immediately above the likely cloud.
6. What temperature feature between 10 and 12 km suggests the tropopause has been reached?

YOU SHOULD TRY Question 3: Quantify the vertical wind shear

Between the surface and 10 km:

1. By how much does wind speed increase?
2. What is the mean increase in wind speed per kilometre over this depth?
3. By how many degrees does the reported wind direction change, and does it veer clockwise or back anticlockwise with height?
4. State one forecasting reason why this vertical change in wind cannot be inferred from surface observations alone.

EXTRA CHALLENGE Question 4: Why does the balloon expand and burst?

A weather balloon has volume 2.0 m^3 at launch, where $p_1 = 1010 \text{ hPa}$ and $T_1 = 287 \text{ K}$. Near the top of the profile, take $p_2 = 200 \text{ hPa}$ and $T_2 = 220 \text{ K}$. Neglect gas leakage and assume ideal-gas behaviour.

1. Calculate the balloon volume at the upper level using $pV/T = \text{constant}$.
2. By what factor has its volume increased?
3. If the envelope fails at about 8.0 m^3 , is a burst near this level plausible?

EXTRA CHALLENGE Question 5: Estimate ascent time and horizontal drift

Suppose the balloon ascends at a constant 5.0 m s^{-1} to 10 km. For a rough drift estimate, take the layer-mean horizontal wind speed to be 20 m s^{-1} .

1. How long does the ascent to 10 km take in seconds and minutes?
2. How far horizontally would the balloon travel at the assumed mean wind speed?
3. Give two reasons why the real trajectory would differ from this straight-line estimate.

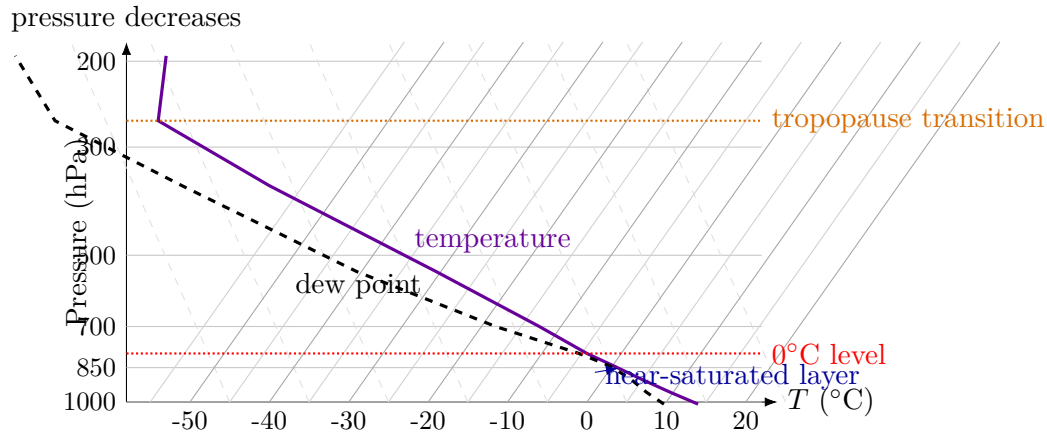
YOU SHOULD TRY Question 6: Use the sounding for an operational forecast

A precipitation system is approaching the launch site. Use only the sounding information to assess the following statements as *supported*, *not supported*, or *cannot be decided from the sounding alone*. Give a brief reason for each.

1. Low cloud is likely between about 1 and 2 km.
2. Any snow generated above 3 km must reach the surface as snow.
3. Aircraft could encounter a freezing level near 2 km.
4. Stronger winds occur aloft than at the surface.
5. The sounding proves that heavy rain will occur during the next hour.

IN CLASS Question 7: Read a plotted sounding rather than a table

The same synthetic sounding is plotted below on a simplified tephigram-style thermodynamic diagram. The solid purple curve is temperature and the dashed black curve is dew point. Grey guide lines show isotherms, dry adiabats and pressure levels. Use the diagram together with the table.



1. Identify the likely cloud layer from the separation between temperature and dew point.
2. Estimate the freezing level and tropopause height.
3. Where is the driest lower/mid-tropospheric air evident?
4. Explain one advantage of seeing the profile graphically rather than only as a table of numbers.